Maloney's Geomethry

An Axiomatic Formalization of Synthetic Geometry

$$\mathfrak{M} \coloneqq \langle \mathbf{P}, \mathcal{B}, \cong, \mathcal{L}, \mathcal{A} \rangle$$

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Contents

Guide to Notation

Objects and Sets

A, B, C, $\ldots$, X, Y, Z
Capital italic: **points**
$\mathbf{P}$ The set of all points
$\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, $\ldots \mathfrak{X}$, $\mathfrak{Y}$, $\mathfrak{Z}$ Capital x: geometric figures
$\overline{AB}$ **Line Segment**
$\overleftrightarrow{AB}$ **Line**
$\overrightarrow{AB}$ **Ray**
$\triangle ABC$ **Triangle**
$\angle ABC$ **Angle**
$\overline{AB} \parallel \overline{CD}$ **Parallel**
$\overline{AB} \perp \overline{CD}$ **Perpendicular**

0.1 Primative Relations

$\mathcal{B}(A, B, C)$ Betweeness
$A = B$ Equivalence
$\overline{AB}$, $\overline{CD}$ Congruence

0.2 idk

$\mathcal{L}$ The **Language**
$\mathcal{A}$ The **Axiom Set**

Chapter 1

Primitive Notions

To build a deductive theory of geometry we must first fix the very language in which the theory will be expressed. Like any mathematical discipline, geometry begins with a collection of *primitive* concepts - terms that we do not define in terms of anything simpler but whose mutual relationships are governed by a system of axioms. From these axioms, we will derive theorems regarding the behavior and properties of these primative notions. For our axiomatic system we adopt three primitive notions: one kind of object, and two relations among those objects. Together they form a single structure that captures the intuitive idea of a flat plane, and will later be used to prove represents a flat plane given our chosen axioms.

The first primitive notion is the **point**. Points are the individuals of our theory; they are the entities that occupy locations in the plane without themselves having parts or extension.

On this set we impose two primitive relations. The first is a *ternary* relation $\mathcal{B}$ ("betweenness"), and the second is a *quaternary* relation $\cong$ ("congruence"). These three ingredients - the point-set and the two relations - are the sole raw material from which the whole theory will be constructed.

The entire structure, consisting of the domain $\mathbf{P}$ equipped with the relations $\mathcal{B}$ and $\cong$, is what we call the **plane** and denote by $\mathcal{P}$.

1.1 Plane

Before dissecting the individual notions, it is useful to definine the whole apparatus formallys.

Definition 1 (Plane)**.** *A plane is a structure*

$$\mathcal{P} := \langle \mathbf{P}, \mathcal{B}, \cong \rangle,$$

b where $\mathbf{P}$ *is an arbitrary set,* $\mathcal{B}$ *is a ternary relation on* $\mathbf{P}$*, and* $\cong$ *is a quaternary relation on* $\mathbf{P}$*.*

Thus, saying "let $\mathcal{P}$ be a plane" amounts to specifying a set of points together with a particular interpretation of betweenness and congruence satisfying the axioms we shall gradually introduce. The definition deliberately uses the neutral term "a plane" rather than "the Euclidean plane" because, until we make further choices, the axioms we give will be true in a broad class of geometries.

1.2 Points

The elements of the domain $\mathbf{P}$ are termed *points*. We denote points by upper-case Latin letters: $A, B, C, \ldots$. All points are gathered into a set $\mathbf{P}$, called the *point-set*, defined by

$$\mathbf{P} := \{\, P \mid P \text{ is a point} \,\}.$$

Within the formal system, a point is an object whose only properties are those granted to it by the axioms governing $\mathcal{B}$ and $\cong$. There is no further internal structure: a point is defined not by decomposition into simpler parts, but solely by its membership in the set $\mathbf{P}$.

Axiom 1 (Existence of Distinct Points)**.** $\exists A, B \in \mathbf{P}, A \neq B$.

This axiom guards against the trivial one-point model satisfying every later axiom.

Informal characterization. Although the formal theory treats points as primitives, it is helpful to keep in mind the intuitive picture that guides our choice of axioms. A point can be thought of as a unique, indivisible position in the plane. This corresponds to the classical Euclidean characterization of a point as "that which has no part." Informally, a point is regarded as having no magnitude - no length, width, or depth - and as being zero-dimensional. However, this physical description is only a heuristic. Within the deductive framework, the formal behaviour of points is entirely determined by the relations in which they stand, and the intuitive characterization carries no logical weight in the proofs that follow. We draw on the geometric imagery only to motivate the axioms and to make the abstract relations vivid.

1.3 Betweenness

Betweenness is a ternary relation on the set of all points, denoted by $\mathcal{B} \subseteq \mathbf{P}^3$. Given points A, B, and C, we write $\mathcal{B}(A, B, C)$ and read it as "B lies between A and C".

Unlike a set $\{A, B, C\}$, where the order of elements is immaterial, betweenness is defined over the cartesian product $\mathbf{P} \times \mathbf{P} \times \mathbf{P}$. Thus it depends on the order of the arguments. Formally,

$$\mathcal{B} \subseteq \mathbf{P}^3 := \{(A, B, C) \mid A, B, C \in \mathbf{P}\}.$$

The expression $\mathcal{B}(A, B, C)$ is simply the assertion that $(A, B, C) \in \mathcal{B}$.

In Euclidean geometry a point B is between A and C precisely when the three points lie on a line and B is met while traveling from A to C along that line.Our axioms capture this idea without invoking lines or order directly.

We now x five axioms governing the behavior of $\mathcal{B}$.

1.3.1 Axioms of Betweenness

Axiom 2 (Symmetry of Betweenness)**.**

$$\forall A, B, C \in \mathbf{P}, \mathcal{B}(A, B, C) \implies \mathcal{B}(C, B, A).$$

Betweenness is symmetric with respect to the outer points: if B lies between A and C, then it also lies between C and A. This reflects the fact that the segment AC is the same as the segment CA. Left-to-right implication is used over a biconditional due to redundancy, as the right-to-left implication directly follows from this form of the axiom.

Axiom 3 (Order)**.**

$$\forall A, B, X \in \mathbf{P},\ \mathcal{B}(A, B, X) \implies \big(\neg\mathcal{B}(B, A, X) \ \wedge\ \neg\mathcal{B}(A, X, B)\big).$$

If B is between A and X, then A cannot be between B and X, nor can X be between A and B. In other words, a point can serve as a "middle" in only one way: the triple (A, B, X) determines a unique order along the line. Together with the Identity axiom, this implies that

$\mathcal{B}(A, A, A)$ is impossible (otherwise, taking $A = B = X$, we would have both $\mathcal{B}(A, A, A)$ and its negation). Thus the relation is *non-reflexive* in the strict sense: no point lies between itself and itself.

Axiom 4 (Identity of Betweenness)**.**

$$\forall A, B \in \mathbf{P}, \mathcal{B}(A, B, A) \implies A = B.$$

This axiom says that if a point appears as the "middle" of a triple that begins and ends with the same point, then the two outer points must coincide. In particular, it forbids the possibility that a distinct point B B lies between A and A; such a configuration would contradict the intuition that betweenness requires a genuine interval. Notice that the converse, $A = B \implies \mathcal{B}(A, A, A)$, is *not* assumed; indeed the next axiom will exclude it.

Axiom 5 (Density)**.**

$$\forall A, C \in \mathbf{P}, A \neq C \implies \exists B \in \mathbf{P} \colon \mathcal{B}(A, B, C).$$

Whenever two points are distinct, there exists a point strictly between them. This captures the idea that segments are dense—between any two points one can always find a third.

The plane must be infinite. Thus, we introduce an axiom that will later allow us to guarenteed the existence of infinite points along a line.

Axiom 6 (Extensibility)**.** *Given two points A and B, there exists a point C such that* $\mathcal{B}(A, B, C)$

$$\forall A, B \in \mathbf{P}, A \neq B \implies \exists C \in \mathbf{P} \colon \mathcal{B}(A, B, C).$$

Axiom 7 (Transitivity of Betweenness)**.** $\mathcal{B}(A, B, C) \wedge \mathcal{B}(B, C, D) \wedge B \neq C \implies \mathcal{B}(A, B, D)$.

If B lies between A and C, and C lies between B and D (with $B \neq C$), then B lies between A and D. This axiom allows us to chain betweenness relations along a line.

These six axioms are mutually independent: no one can be derived from the others (as can be shown by standard model-theoretic arguments). Together with the congruence axioms to be introduced later, they will form the basis of our geometric theory.

1.3.2 Immediate Consequences

As is, the extensibility axiom allows only for extensibility to the right. To prove this consisely for both directions we introduce a lemma demostrating extensibility on the left side:

Theorem 1 (Left-Sided Extensibility)**.** *For any two points A and B, there exists C such that $\mathcal{B}(C, B, A)$.*

Proof. From $A \neq B$, the extensibility axiom 6 implies $\mathcal{B}(A, B, C)$. We then apply the symmetry of betweenness 2 to obtain $\mathcal{B}(C, B, A)$, thus arriving at the desired result. □

With this, we can now prove extensibility on both sides.

Theorem 2 (Dual-Sided Extensibility)**.** *Given any two points B and C, there exists points A and D such that $[ABCD]$.*

Proof. Let $B, C \in \mathbf{P}$ be two points not equal to each other. From the extensibility axiom. there exists a point D such that $[BCD]$. Similarily, from left-sided extensibility, there exists a point A such that $[ABC]$. These results are sufficent to demostrate that $[ABCD]$, thus proving the theorem. □

Theorem 3. $\mathcal{B}(A, B, C) \iff \mathcal{B}(C, B, A)$ *for all* $A, B, C \in \mathbf{P}$.

Proof. The forward direction is the axiom; the reverse follows by applying the axiom with A and C exchanged. □

Given two distinct points, there is always a point beyond the second, so that the second lies between the first and the new point. This guarantees that lines extend indefinitely.

These five axioms are mutually independent: no one can be derived from the others (as can be shown by standard model-theoretic arguments). Together with the congruence axioms to be introduced later, they will form the basis of our geometric theory.

Theorem 4 (Irreflexivity of Betweenness)**.** *No point lies between it and itself:*

$$\forall A \in \mathbf{P},\ \neg\mathcal{B}(A, A, A).$$

Proof. Assume $\mathcal{B}(A, A, A)$. By the order axiom, this implies $\neg\mathcal{B}(A, A, A)$ and $\neg\mathcal{B}(A, A, A)$, which contradicts our assumption. Hence, $\neg\mathcal{B}(A, A, A)$. □

Theorem 5 (Distinct Endpoints)**.** *Betweenness forces distinct outer points:*

$$\mathcal{B}(A, B, C) \implies A \neq C.$$

Proof. Suppose $\mathcal{B}(A, B, C)$ and $A = C$. Then, we have $\mathcal{B}(A, B, A)$, which by the identity of betweenness implies $A = B$. Substituting back gives $\mathcal{B}(A, A, A)$, which contradicts the irreflexivity of betweenness. Therefore $A \neq C$. □

Theorem 6 (Irreflexivity of Betweenness for Distinct Points)**.**

$$A \neq B \implies \neg\mathcal{B}(A, A, B) \wedge \neg\mathcal{B}(A, B, B).$$

Proof. If $\mathcal{B}(A, A, B)$, then by the order of betweeness, □

Theorem 7 (Strict Betweenness)**.** *If* $\mathcal{B}(A, B, C)$*, then* $A \neq B$ *and* $B \neq C$*.*

Proof. Suppose $\mathcal{B}(A, B, C)$ and $A = B$. Then, $\mathcal{B}(A, A, C)$, which implies $\neg\mathcal{B}(A, A, C)$ by the order of betweenness. However, this is a contradiction. Thus, $A \neq B$. Similarly, if we assume $B = C$, then we have $\mathcal{B}(A, B, B) \implies \neg\mathcal{B}(A, B, B)$, leading to a contridiction. Thus, $B \neq C$. □

Theorem 8 (Contrapositive Ordering)**.** *For all* $A, B, X \in \mathbf{P}$*,*

$$\mathcal{B}(B, A, X) \implies \neg\mathcal{B}(A, B, X) \quad \textit{and} \quad \mathcal{B}(A, X, B) \implies \neg\mathcal{B}(A, B, X).$$

Proof. The Order axiom gives the two implications

$$\mathcal{B}(A, B, X) \implies \neg\mathcal{B}(B, A, X) \quad \text{and} \quad \mathcal{B}(A, B, X) \implies \neg\mathcal{B}(A, X, B).$$

Taking the logical contrapositive of each implication(Iwhich is logically equivalent to the original), yields precisely the two stated results. □

Theorem 9 (Symmetry Extended)**.** *Given* $\mathcal{B}(A, B, C)$ *none of the following hold:* $\mathcal{B}(B, A, C)$*,* $\mathcal{B}(A, C, B)$*,* $\mathcal{B}(B, C, A)$*,* $\mathcal{B}(C, A, B)$*.*

These theorems complete our baseline analysis of how the betweenness relation $\mathcal{B}$ operates locally on individual triples of points. However, proving that the plane $\mathcal{P}$ is structurally infinite—both in its internal density and its outward extension—requires the formal definitions of segments and lines. We defer these proofs to chapter blank, where the infinite nature of the geometric line can be properly established.

1.4 Congruence

Congruence is a quaternary relation on the set of all points denoted by $\cong \subseteq \mathbf{P}^4$. Rather than mapping individual points to one another, this relation establishes a geometric equivalence between two pairs of points. We write $\cong (A, B, C, D)$, to assert that the segment from point A to point B is congruent to the segment from point C to point D. Formally:

$$\cong (A, B, C, D) \iff (A, B, C, D) \in \cong .$$

To reflect this conceptually, we bundle the point pairs (A, B) and (C, D) into the typographical units AB and CD and introduce the infix symbol $\cong$:

$$AB \cong CD \overset{\text{def}}{\iff} \cong (A, B, C, D)$$

If the relation holds for the pairs (A, B) and (C, D), we write $(A, B) \cong (C, D)$, which is interpreted as the "distance" from A to B being equal to the distance from C to D. The notation for the points will be reused to define segments. This choice was intentional, as the later informal usage of congruence upon segments is identical in nature to what is presented below.

We now impose three axioms governing the behavior of $\cong$.

1.4.1 Axioms of Congruence

Axiom 8 (Reversal of Congruence)**.**

$$\forall A, B \in \mathbf{P},\ AB \cong BA.$$

Axiom 9 (Transitivity of Congruence)**.**

$$\forall A, B, C, D, E, F \in \mathbf{P},\ \big(AB \cong CD \wedge AB \cong EF\big) \implies CD \cong EF.$$

Transitivity (Axiom 9) is the formal way of saying that two segments congruent to the same segment are congruent to each other.

With only Reversal and Transitivity, however, the axioms do not distinguish between genuine segments (whose endpoints are distinct) and degenerate ones (whose endpoints coincide). Model in which every point pairs is congruent to one another would strictly satifsy the above. In this universe, distinct points exist (Axiom 1), but they are not separated. Such a "zero-distance" universe reduces the congruence relation to the universal relation and makes the geometry collapse: all points are at the same distance from one another, and no meaningful notion of length can be recovered.

To exclude this degenerate model we introduce the following.

Axiom 10 (Identity of Congruence)**.**

$$\forall A, B, C \in \mathbf{P},\ AB \cong CC \implies A = B.$$

In words: if a segment is congruent to a degenerate segment, then its endpoints must coincide. Equivalently, a non-degenerate segment ($A \neq B$) can never be congruent to a degenerate one (CC). Intuitively, this means the "distance" between distinct points is never zero, — a requirement that, in a metric geometry, would be imposed by the condition $d(A, B) = 0 \iff A = B$.

Note that the axiom does not guarenteed the existence of distinct points — that guarantee was already given by the domain axiom Axiom 1. However, these axioms together guarantee that the plane contains at least two points *and* that distinct points are geometrically distinct.

Once distinct, genuinely separated points exist, **??** populates the segment between them with infinitely many intermediate points, ensuring that the line is not a discrete set of isolated locations.

1.4.2 Congruence as an Equivalence Relation

An equivalence relation is characterized by three properties: reflexivity, symmetry, and transitivity. The axioms adopted here for $\cong$ assert neither reflexivity nor the general symmetry and transitivity required of an equivalence relation directly; Transitivity, in particular, is stated in a shared-first-argument form rather than the familiar chain form. The remainder of this subsection shows that Reversal and Transitivity nonetheless force Reflexivity, Reflexivity together with Transitivity forces Symmetry, and Symmetry together with Transitivity forces the familiar chain form of transitivity, so nothing further needs to be assumed.

Theorem 10 (Reflexivity of Congruence)**.**

$$\forall A, B \in \mathbf{P},\ AB \cong AB.$$

Proof. From reversal (Axiom 8), we have $AB \cong BA$. Through replacement of variables we also have $BA \cong AB$. We now instantiate transivity (Axiom 9) with $AB \cong BA$ used twice:

$$\big(AB \cong BA \text{ and } AB \cong BA\big) \implies BA \cong BA.$$

Next, we apply transivity again as follows:

$$\big(BA \cong AB \text{ and } BA \cong AB\big) \implies AB \cong AB,$$

which completes the proof. □

With reflexivity established, the other expected properties follow rapidly.

Theorem 11 (Symmetry of Congruence)**.** *If $AB \cong CD$, then $CD \cong AB$.*

Proof. We are given $AB \cong CD$. From the reflexivity axiom we also have $AB \cong AB$. Applying these two congruences to the transitivity axiom gives the antecedent

$$AB \cong CD \wedge AB \cong AB,$$

which implies $CD \cong AB$. □

Theorem 12 (Full-Transitivity of Congruence)**.** *If $AB \cong CD$ and $CD \cong EF$, then $AB \cong EF$.*

Proof. By hypothesis, $AB \cong CD$. Applying Symmetry (Theorem 11) to this gives

$$CD \cong AB.$$

Together with the second hypothesis $CD \cong EF$, we obtain the conjunction

$$CD \cong AB \ \wedge \ CD \cong EF.$$

This is precisely an instance of the Transitivity axiom (Axiom 9), instantiated with CD in the shared first-argument position of both conjuncts. The axiom therefore yields

$$AB \cong EF,$$

as required. □

With these results, $\cong$ is established as an equivalence relation on ordered pairs of points: reflexive and symmetric by theorem, and transitive in the familiar chain sense by Full-Transitivity. Consequently, $\cong$ partitions the set of all ordered point pairs into disjoint equivalence classes, each consisting of all segments congruent to one another — the geometric counterpart of segments of equal length. This is the extent of what an equivalence relation alone provides: it identifies which segments0 share a length, but supplies no order between distinct classes and no arithmetic operation upon them. Such structure requires the interaction of $\cong$ with $\mathcal{B}$ and is developed in a later section.

1.4.3 Endpoint-Reversal Symmetries

The equivalence-relation properties established above govern the behavior of $\cong$ across distinct pairs of points; they say nothing about the effect of reordering the two endpoints *within* a single pair. This subsection establishes that $\cong$ is in fact invariant under such reordering, in every combination.

Theorem 13 (Left-Reversal of Congruence)**.** *If $AB \cong CD$, then $BA \cong CD$.*

Proof. We are given $AB \cong CD$. From the reversal axiom, we also have $AB \cong BA$. Applying transitivity to the conjunction

$$AB \cong BA \wedge AB \cong CD$$

yields $BA \cong CD$. □

Theorem 14 (Right-Reversal of Congruence)**.** *If $AB \cong CD$, then $AB \cong DC$.*

Proof. We are given $AB \cong CD$. From the reversal axiom applied to the pair (C, D), we have $CD \cong DC$. By symmetry (Theorem 11) applied to the hypothesis, we also have $CD \cong AB$. Applying transitivity to the conjunction

$$CD \cong AB \wedge CD \cong DC,$$

yields $AB \cong DC$. □

Theorem 15 (Full-Reversal of Congruence)**.** *If $AB \cong CD$, then $BA \cong DC$.*

Proof. We are given $AB \cong CD$. From the reversal axiom, we know that $AB \cong BA$. By Right-Reversal (Theorem 14) applied to the hypothesis, we also have $AB \cong DC$. Applying transitivity to

$$AB \cong BA \wedge AB \cong DC,$$

yields $BA \cong DC$. □

The previous three theorems can be packaged into a single convenient equivalence, establishing that segment congruence is entirely indifferent to the order of the endpointsof either segment.

Theorem 16 (Endpoint-Swap Invariance)**.** *For any points A, B, C, D,*

$$AB \cong CD \iff BA \cong DC \iff AB \cong DC \iff BA \cong CD.$$

Proof. The forward implications have all been proved:

- $AB \cong CD \Rightarrow BA \cong DC$ (Full-Reversal)
- $BA \cong DC \Rightarrow AB \cong DC$ (apply Left-Reversal to the pair (B, A))
- $AB \cong DC \Rightarrow BA \cong CD$ (Right-Reversal followed by Left-Reversal)
- $BA \cong CD \Rightarrow AB \cong CD$ (Left-Reversal on (B, A))

The reverse directions follow by the same theorems applied in the opposite order, completing the circle of equivalences. □

1.5 Interaction of Betweenness and Congruence

Axiom 11 (Segment Construction)**.**

$$\forall Q, X, A, B \in \mathbf{P},\ Q \neq X \implies \exists Y \in \mathbf{P} \colon \mathcal{B}(Q, X, Y) \wedge XY \cong AB.$$

Chapter 2

Linear Notions

Now that the domain and relations of our plane have been fleshed-out, we set off to define objects present on the plane. That is, sets of points that meeting a defined criteria. Objects like this are called *figures*, wherein a figure F on the plane is a subset of the point set: $F \subseteq \mathbf{P}$. Note that a figure is a proper subset of $\mathbf{P}$, as the set containing no points is simply nothing, and the set containing all points is the domain itself.

The most fundamental types of geometric figures are the segment, ray, and line - which we classifify as *linear notions*. In this chaoter, we define all three linear notions, and detirmine their basic properties.

Theorem 17 (Infinite interior points)**.** *Let $A, C \in \mathbf{P}$ with $A \neq C$. Then there exist infinitely many points strictly between A and C.*

Proof. wasd □

Theorem 18 (Infinite outward extension)**.** *Let $A, B \in \mathbf{P}$ with $A \neq B$. Then there exist infinitely many points $C_1, C_2, C_3, \ldots$ such that*

$$\mathcal{B}(A, B, C_1), \quad \mathcal{B}(A, C_1, C_2), \quad \mathcal{B}(A, C_2, C_3), \quad \ldots$$

Proof. wasd □

Together, these two lemmas show that from any segment AB ($A \neq B$) we can generate infinitely many points in both directions along the line determined by A and B. Thus the plane $\mathcal{P}$ is necessarily infinite.

2.1 Segments

A *line segment* AB is the set containing A, B, and all points between A and B. The points A and B are called the *endpoints* of the segment. Symbolically, that is

Definition 2 (Segment)**.**

$$\forall A, B \in \mathbf{P}, A \neq B \implies AB := \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B)\}.$$

By intuition, we can see that the segment AB is identical to the segment BA. Using properties of sets upon the endpoints amd the symmetry of betweeness 2 upon the point in between X we can prove this result.

Theorem 19 (Segment Symmetry)**.** *For any two points A, B, the segment AB is equal and equivalent to the segment BA.*

$$\forall A, B \in \mathbf{P}, AB = BA.$$

Proof. By the definition of a line segment, we have:

$$AB = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B)\}$$

and

$$BA = \{B, A\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(B, X, A)\}.$$

First, by the commutative property of sets, the endpoints are identical: $\{A, B\} = \{B, A\}$.

Second, we examine the sets of interior points. By the **Symmetry of Betweenness**, the relation $\mathcal{B}(A, X, B)$ holds if and only if $\mathcal{B}(B, X, A)$ holds. Formally, this implies:

$$\{X \in \mathbf{P} \mid \mathcal{B}(A, X, B)\} = \{X \in \mathbf{P} \mid \mathcal{B}(B, X, A)\}.$$

Since AB and BA are defined as the union of identical sets, they contain the same points. By the principle of extensionality in set theory, $AB = BA$. □

2.2 Rays

We now consider the case where a line segment extends out in one direction. A ray pointing from point A to a discint point B is a subset of $\mathbf{P}$ denoted as $\overrightarrow{AB}$. It is defined as the union of A, B, all points between A and B, and all points such that B is between A and that point, that is

Definition 3 (Ray)**.**

$$\forall A, B \in \mathbf{P}, A \neq B \implies \overrightarrow{AB} \coloneqq \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\}.$$

The point the ray begins at is called the idk. Our notation will allow for the arrow to be facing left or right. However, it is preferred to have the arrow pointing right.

Theorem 20 (Ray Asymmetry)**.** *For any two distinct points $A, B \in \mathbf{P}$, the ray $\overrightarrow{AB}$ is not equal to the ray $\overleftarrow{BA}$:*

$$\forall A, B \in \mathbf{P}, A \neq B \implies \overrightarrow{AB} \neq \overrightarrow{BA}.$$

Proof. We proceed by by contradiction. Assume that the rays for some distinct points $A, B \in \mathbf{P}$, $\overrightarrow{AB} = \overrightarrow{BA}$.

By the definition of a ray, we have:

$$\overrightarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\},$$

and

$$\overrightarrow{BA} = \{B, A\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(B, X, A) \vee \mathcal{B}(B, A, X)\}.$$

By the **Outer Point Axiom**, there must exist a point Y such that $\mathcal{B}(A, B, Y)$. By the defintion of $\overrightarrow{AB}$, it follows that $Y \in \overrightarrow{AB}$. Under our initial assumption that the rays, Y must be an element of $\overrightarrow{BA}$.

By the Distinctness of Betweeness, the relation $\mathcal{B}(A, B, Y)$ implies that $A \neq Y$ and $B \neq Y$. Therefore, $Y \notin \{A, B\}$.

Since $Y \in \overrightarrow{BA}$ but is not an endpoint, it must satisfy $\mathcal{B}(B, Y, A)$ or $\mathcal{B}(B, A, Y)$.

However, because we know that $\mathcal{B}(A, B, Y)$ is true, by Order of Betweeness we have $\mathcal{B}(A, B, Y) \implies \neg\mathcal{B}(B, A, Y)$, and $\mathcal{B}(A, B, Y) \implies \neg\mathcal{B}(A, Y, B)$, which by the Symmetry of Betweeness is logically equivalent to $\mathcal{B}(A, B, Y) \implies \neg\mathcal{B}(B, Y, A)$.

Thus, both conditions for Y to be an element of $\overrightarrow{BA}$ are false. Therefore, the rays $\overrightarrow{AB}$ and $\overrightarrow{BA}$ are not equal, thus proving the desired result. □

Theorem 21 (Union of Opposite Rays)**.** *For any two points $A, B \in \mathbf{P}$,*

$$\overleftarrow{AB} \cup \overrightarrow{AB} = \overleftrightarrow{AB}.$$

Proof. By the definition of a ray, we have:

$$\overleftarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(X, A, B) \vee \mathcal{B}(A, X, B)\},$$

and

$$\overrightarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\}.$$

The union of $\overleftarrow{AB}$ and $\overrightarrow{AB}$ therefore consists of the set, $\{B, A\}$, and all points X such that $\mathcal{B}(X, A, B)$ or $\mathcal{B}(A, X, B)$, or $\mathcal{B}(A, B, X)$.

Note that this is is the same as the line $\overleftrightarrow{AB}$, as it is defined as the set

$$\overleftrightarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(X, A, B) \vee \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\}.$$

This set is equal to the union described above. Thus, the union of the rays $\overleftarrow{AB}$ and $\overrightarrow{AB}$ is equal to the line $\overleftrightarrow{AB}$. □

Theorem 22 (Intersection of Opposite Rays)**.** *For any two points $A, B \in \mathbf{P}$,*

$$\overleftarrow{AB} \cap \overrightarrow{AB} = AB.$$

Proof. By the definition of a ray, we have:

$$\overleftarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(X, A, B) \vee \mathcal{B}(A, X, B)\},$$

and

$$\overrightarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\}.$$

The intersection of $\overleftarrow{AB}$ and $\overrightarrow{AB}$ therefore consists of the set, $\{B, A\}$ and all points X betweem A and B, that is $\mathcal{B}(A, X, B)$.

This intersection consists of the same points contained in the segment AB, as

$$AB = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B)\}.$$

Thus, the union of the rays $\overleftarrow{AB}$ and $\overrightarrow{AB}$ is equal to the segment AB. □

2.3 Lines

The line $\overleftrightarrow{AB}$ is the subset of $\mathbf{P}$ consisting of the points A, B, all points between A and B, all points where A is between it and B, and all points where B is between it and A. That is,

Definition 4 (Line)**.**

$$\forall A, B \in \mathbf{P}, A \neq B \implies \overleftrightarrow{AB} := \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(X, A, B) \vee \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\}.$$

Theorem 23 (Line Symmetry)**.** *For any two distinct points A and B, the line $\overleftrightarrow{AB}$ is equal to the segment $\overleftrightarrow{BA}$:*

$$\forall A, B \in \mathbf{P}, \overleftrightarrow{AB} = \overleftrightarrow{BA}.$$

Proof. By the definition of a line, we have:

$$\overleftrightarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(X, A, B) \vee \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\}$$

and

$$\overleftrightarrow{BA} = \{B, A\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(X, B, A) \vee \mathcal{B}(B, X, A) \vee \mathcal{B}(B, A, X)\}.$$

First, by the commutative property of sets, the endpoints are identical: $\{A, B\} = \{B, A\}$.

Second, we demonstrate that the three cases of betweeness in $\overleftrightarrow{AB}$ are logically equivalent to those of $\overleftrightarrow{BA}$ using the by the **Symmetry of Betweenness**:

- $\{X \in \mathbf{P} \mid \mathcal{B}(X, A, B)\} = \{X \in \mathbf{P} \mid \mathcal{B}(B, A, X)\}$
- $\{X \in \mathbf{P} \mid \mathcal{B}(A, X, B)\} = \{X \in \mathbf{P} \mid \mathcal{B}(B, X, A)\}$
- $\{X \in \mathbf{P} \mid \mathcal{B}(A, B, X)\} = \{X \in \mathbf{P} \mid \mathcal{B}(X, B, A)\}$

Since each condition for membership in $\overleftrightarrow{AB}$ is logically equivalent to a condition for membership in $\overleftrightarrow{BA}$, the sets contain the same points. Thus, by the principle of extensionality, $\overleftrightarrow{AB} = \overleftrightarrow{BA}$ □

2.4 Midpoints and Bisectors

Intuitively, the midpoint of a segment is its "center" - the point that splits the segment into two equal halves. In real life, you find a midpoint when you fold a piece of paper so that the ends meet; the crease marks the exact middle. But in our geometry, which is built only on points, betweenness, and congruence, we need to be precise: a point is the midpoint of segment AB exactly when it lies on the segment (so it is between A and B) and the two subsegments it creates are congruent.

THowever, we must ensure that such a point always exists and that there is only one. Geometry would be very strange if some segments could not be cut in half, or if a segment had two distinct "centers." We introduce two axioms to gaurenteed this, but first present a formal definition of a midpoint.

Definition 5 (Midpoint)**.** *For distinct points $A, B \in \mathbf{P}$, a point M is a midpoint of segment AB if*

$$\mathcal{B}(A, M, B) \quad \textit{and} \quad AM \cong MB.$$

Now, the axioms:

Axiom 12 (Midpoint Existence)**.** $\forall A, B \in \mathbf{P},\ A \neq B \implies \exists M \in \mathbf{P} : \big(\mathcal{B}(A, M, B)\big) \wedge \big(AM \cong MB\big).$

Axiom 13 (Midpoint Uniqueness)**.** $\forall A, B \in \mathbf{P},\ A \neq B \implies \forall M, N \in \mathbf{P} : \big(AM \cong MB \wedge \mathcal{B}(A, M, B) \wedge AN \cong NB \wedge \mathcal{B}(A, N, B)\big) \implies M = N.$

The first ensures that given two points A and B, there exists a point M which is the midpoint of the segment AB. However, this alone allows for geometeries in which segments have multiple midpoints. Thus, we introduce an axiom ensuring the uniqueness of the midpoint stating that for segment AB, if points M and N both meet the crieteria for being a midpoint, then it is nessessary that $M = N$. These axioms therefore require existence of a unique midpoint given any line segment.

Now we ask: what happens if we drop the requirement that the point lies on the segment, but keep the requirement for congruence. That is, we detirmine all points X such that $AX \cong XB$. In general, a line containing the midpoint of a segment is called a bisector of that segment. However, this set of points gets its own name.

Definition 6 (Perpendicular Bisector)**.** *For distinct points $A, B \in \mathbf{P}$, the perpendicular bisector of segment AB is the set*

$$\ell_{AB} \coloneqq \{\, X \in \mathbf{P} \mid AX \cong XB \,\}.$$

The name is justified by two facts (which we will prove later):

1. ℓ_{AB} is a line (i.e., there exist points P, Q such that $\ell_{AB} = \overleftrightarrow{PQ}$);
2. ℓ_{AB} is perpendicular to the line $\overleftrightarrow{AB}$ and passes through the unique midpoint M of AB.

Notice that the midpoint M itself belongs to ℓ_{AB} because $AM \cong MB$, and M is the only point of ℓ_{AB} that lies on the segment AB (the betweenness condition forces it to be between A and B). The rest of the perpendicular bisector lies off the segment, forming the two rays that are perpendicular to AB at M.

Remark. Defining the perpendicular bisector as the equidistant set $\{\, X \mid AX \cong XB \,\}$ has the advantage of making no prior reference to perpendicularity or to the midpoint; both properties are then *derived* from the congruence and betweenness axioms. This is the standard treatment in absolute (neutral) geometry.

2.5 Parallelism

Having introduced the fundamental linear figures—segments, rays, and lines—we now turn our attention to a relation that captures one of the most essential geometric intuitions: the notion that two lines "point in the same direction" and therefore never meet. This relation is called *parallelism*. In a purely set-theoretic framework, where lines are subsets of the point set $\mathbf{P}$, the idea that two distinct lines have no point in common is naturally expressed as the emptiness of their intersection. There is, however, a subtlety concerning the limiting case in which the two lines are actually the same set: should a line be considered parallel to itself? Different axiomatic traditions answer this differently. In the ordinary language of school geometry one often says that parallel lines are distinct lines that do not intersect, thereby excluding the reflexive case. In a formal development, however, it is far more convenient to adopt the inclusive convention and allow a line to be parallel to itself. Doing so makes parallelism an equivalence relation—a structure that will prove invaluable later when we classify lines by their *direction*. The inclusive definition is the one we adopt here; it simplifies the statement of theorems.

Because parallelism is a relation that compares lines, we must first collect all lines into a single object that can serve as the domain of the relation. Our definition of a line $\overleftrightarrow{AB}$ yields a subset of $\mathbf{P}$ for every pair of distinct points A, B; the collection of all such subsets is therefore a well-defined family of sets. We denote this family by $\mathbf{L}$ and make it the official set of lines of our plane.

Definition 7 (Set of all lines)**.** *The set of all lines* $\mathbf{L}$ *is defined as*

$$\mathbf{L} \coloneqq \{\, l \subseteq \mathbf{P} \mid \exists\, A, B \in \mathbf{P},\ A \neq B \wedge l = \overleftrightarrow{AB} \,\}.$$

Every element of $\mathbf{L}$ is a subset of $\mathbf{P}$ that arises from two distinct points by the construction of the line; conversely, every such construction yields a member of $\mathbf{L}$. This definition guarantees that we have a fixed, well-defined set to talk about, a necessary prerequisite for building a binary relation.

Now we are in a position to define parallelism itself. The definition is a direct set-theoretic translation of the geometric idea: two lines are parallel if they coincide or have no points in common.

Definition 8 (Parallel lines)**.** *Let $l, m \in \mathbf{L}$ be lines. We say that l is parallel to m, and write $l \parallel m$, if and only if*

$$l = m \quad \textit{or} \quad l \cap m = \emptyset.$$

Formally,

$$\forall\, l, m \in \mathbf{L}, \qquad l \parallel m \iff (l = m) \;\vee\; (l \cap m = \emptyset).$$

The first disjunct, $l = m$, may at first seem surprising if one is accustomed to thinking of parallel lines as distinct. Its inclusion is deliberate. By allowing a line to be parallel to itself we immediately obtain reflexivity (every line is parallel to itself), and together with the natural symmetry and a suitable parallel postulate we shall obtain transitivity, turning $\parallel$ into an equivalence relation on $\mathbf{L}$. The equivalence classes of this relation are precisely the *directions* in the plane—a concept that lies at the heart of many later arguments. If one preferred the exclusive convention, one could define a separate relation "strictly parallel" and then prove that it differs from $\parallel$ only by missing the diagonal of $\mathbf{L} \times \mathbf{L}$; the two treatments are completely intertranslatable, and we find the inclusive one more elegant.

In the absence of any further axioms, this definition tells us very little about how many parallels exist through a point not on a given line. In fact, neutral (absolute) geometry guarantees the existence of *at least one* such parallel, but leaves the number undetermined. To obtain the familiar Euclidean picture, we shall later introduce an additional postulate—Playfair's Axiom—that asserts the uniqueness of the parallel. Until then, the definition stands as a purely combinatorial condition on subsets of the plane.

Theorem 24 (Reflexivity of Parallelism)**.** *For every line $l \in \mathbf{L}$, $l \parallel l$.*

Proof. Immediate from the definition: take $l = m$ in the disjunct $l = m$. □

Theorem 25 (Symmetry of Parallelism)**.** *For all lines $l, m \in \mathbf{L}$,*

$$l \parallel m \implies m \parallel l.$$

Proof. Assume $l \parallel m$. By definition, either $l = m$ or $l \cap m = \emptyset$. If $l = m$, then $m = l$ and $m \parallel l$ by the same disjunct. If $l \cap m = \emptyset$, then $m \cap l = \emptyset$ because intersection is commutative, so $m \parallel l$. □

Axiom 14 (Playfair's Axiom (Parallel Postulate))**.** *For every line $l \in \mathbf{L}$ and every point $P \in \mathbf{P}$ with $P \notin l$, there exists a unique line $m \in \mathbf{L}$ such that $P \in m$ and $m \parallel l$. Formally,*

$$\forall l \in \mathbf{L},\ \forall P \in \mathbf{P} \setminus l,\ \exists!\, m \in \mathbf{L}\ \big(P \in m \wedge m \parallel l\big).$$

Theorem 26 (Transitivity of Parallelism)**.** *For all lines $l, m, n \in \mathbf{L}$,*

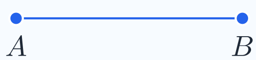
$$(l \parallel m \;\wedge\; m \parallel n) \implies l \parallel n.$$

Proof. Assume $l \parallel m$ and $m \parallel n$. If $l = m$ or $m = n$, the conclusion follows trivially from the given parallelism. Hence we may assume $l \neq m$ and $m \neq n$. Then by definition $l \cap m = \emptyset$ and $m \cap n = \emptyset$.

We must show $l = n$ or $l \cap n = \emptyset$. Suppose, for contradiction, that $l \neq n$ and $l \cap n \neq \emptyset$. Then there exists a point $P \in l \cap n$. Because $l \cap m = \emptyset$, we have $P \notin m$; similarly $P \notin m$ from $m \cap n = \emptyset$. Thus P is a point not on m. Both l and n pass through P and are parallel to m: $l \parallel m$ (given) and $n \parallel m$ (from $m \parallel n$ and symmetry). Playfair's Axiom 14 asserts that there is a *unique* line through P parallel to m. Consequently $l = n$, contradicting the assumption $l \neq n$. Hence our supposition is false; we must have $l \cap n = \emptyset$, and therefore $l \parallel n$. $\square$

Segment AB

$\{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B)\}$

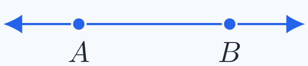


Ray $\overrightarrow{AB}$

Union of AB and points X where $\mathcal{B}(A, B, X)$

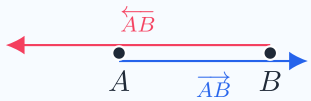
Line $\overleftrightarrow{AB}$

Extends infinitely in both directions



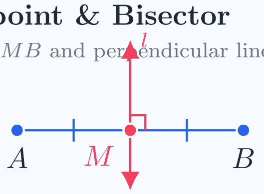
Intersection of Opposite Rays

$\overleftarrow{AB} \cap \overrightarrow{AB} = AB$



Midpoint & Bisector

$AM \cong MB$ and perpendicular line $l \perp AB$



Parallelism ($l \parallel m$)

No points in common: $l \cap m = \emptyset$

Chapter 3

Partition of the Plane

3.1 Plane Separation Axiom

The previous axioms of betweenness govern the linear ordering of points and are therefore limited to a single dimension. To extend this to two dimensions, we introduce the Plane Separation Axiom. This asserts that a line acts as a parition splitting the plane into two discint, non-overlapping regions, which is then used to describe a nototion of points membership to one side of a given line through its logical equivalence to the negation of the previously defined notion of segment intersection.

Axiom 15 (Plane Separation)**.** *For every line $l \in \mathbf{L}$, there exist two sets $H_{l,A}, H_{l,B} \subset \mathbf{P} \setminus l$ such that:*

1. $H_{l,A} \cup H_{l,B} = \mathbf{P} \setminus l \land H_{l,A} \cap H_{l,B} = \emptyset$.
2. $\forall A, B \in \mathbf{P} \setminus l \colon (\exists X \in l, \mathcal{B}(A, X, B) \iff \{A, B\} \not\subset H_{l1} \land \{A, B\} \not\subset H_{l2})$.

explain the meaning-of-axioms-here

3.2 Same-Side Relation

We formalize the concept of 'being on the same side' through the Same-Side Relation ($\mathcal{S}$). It is defined by the absence of a seperation/parition of the plane: two points are related if and only if unique segment connecting them does not contain any point of the boundary line l. This definition ensures the relation is based upon the primative notion of betweeness. Specifically, we define $\mathcal{S}$ as a subset of the cartesian product $\mathcal{S} \subseteq \mathbf{P} \times \mathbf{P} \times \mathbf{L}$.

Definition 9 (Same-Side Relation)**.** *Given a line $l \in \mathbf{L}$, any two points $A, B \in \mathbf{P} \setminus l$ are related by $\mathcal{S}$ if there does not exist a point $X \in l$ such that $\mathcal{B}(A, X, B)$. That is,*

$$\mathcal{S}(A, B, l) \iff \nexists X \in l \colon \mathcal{B}(A, X, B).$$

This geometric object allows us to move beyond comparisons between single points. A point X being on the same same as point Y is the same as X being an element of the half-plane $H_{l,y}$. This will show incredibily useful in defining derived geometric figures.

3.3 Theorems

Theorem 27 (Same-Side Reflexivity)**.** *For any point $A \in \mathbf{P}$ that is not a member of any given line $l \in \mathbf{L}$, the following must hold:*

$$\forall l \in \mathbf{L}, \forall A \in \mathbf{P} \colon \mathcal{S}(A, A, l).$$

Proof. Let there be a line $l \in \mathbf{L}$ and a point $A \in \mathbf{P} \setminus l$. We proceed by contradiction. Assuming the same-side relation $\mathcal{S}(A, A, l)$ is false, it follows that $\nexists X \in l \colon \mathcal{B}(A, X, A)$ is false by definition, and that therefore there does exist an $X \in l$ satisfing these conditions. As a consequence, by the Idenity/Reflexivity of Betweenness $A = X$.

However, we have defined that $A \notin l$. Since our assumption leads to a contradiction, it must be that no such point X exists. Thus, $\mathcal{S}(A, A, l)$ is true. □

Theorem 28 (Same-Side Symmetry)**.**

$$\mathcal{S}(A, B, l) \iff \mathcal{S}(B, A, l).$$

Proof. Firstly, we note that $\mathcal{S}(A, B, l) \iff \nexists X \in l \colon \mathcal{B}(A, X, B)$ by definition. By the Symmetry of Betweeness, the right-hand side of this is equivalent to $\nexists X \in l \colon \mathcal{B}(B, X, A)$, thus showing

$$X \in l \colon \neg\mathcal{B}(A, X, B) \iff X \in l \colon \neg\mathcal{B}(B, X, A).$$

The hand-side is by definition equivalent to $\mathcal{S}(B, A, l)$. Thus, by the transitivity of the biconditional, we conclude $\mathcal{S}(A, B, l) \iff \mathcal{S}(B, A, l)$. □

Theorem 29 (Same-Side Transitivity)**.** *Given any line $l \in \mathbf{L}$ and points $A, B, C \in \mathbf{P} \setminus l$, the relation $\mathcal{S}$ satisfies:*

$$\mathcal{S}(A, B, l) \wedge \mathcal{S}(B, C, l) \implies \mathcal{S}(A, C, l).$$

Proof. By the Plane Seperation Axiom, we know there must exist two distint sets H_{l1} and H_{l2} whose union is $\mathbf{P} \setminus l$. Since A, B, and C are not on l, each must belong to exactly one of these two sets.

Assume $\mathcal{S}(A, B, l)$. By definition, they are contained within the same set. Without loss of generality, let $A, B \in H_{l1}$.

Assume $\mathcal{S}(B, C, l)$. Similarily, this implies they are in the same set. We previously established that $B \in H_{l1}$, implying that $C \in H_{l1}$ by the plane seperation axiom ($H_{l1} \cap H_{l2} = \emptyset$).

We now have $A \in H_{l1}$ and $C \in H_{l1}$. By the inverse of PSA2, there does not exist a point $X \in l$ such that $\mathcal{B}(A, X, C)$. Thus, by definition, A and C must be related by the same-side relation. That is, $\mathcal{S}(A, C, l)$. □

With the reflexivity, symmetry, and transivity of the same-side relation demostrated, it follows by definition that it is an equivalence relation. This result may seem trivial, but it allows for application of known properties of equivlence relations. One such property is *equivalence classes*, which are the subsets for which the relation holds. For our geometry, this partitions the plane into two *half-planes*.

Theorem 30 (Partition Theorem)**.** *Any pair of half-planes alongwith the line forming them together cover the plane.*

Proof. First note that the relation form □

3.4 Half-Planes

By grouping all points related to a reference point A by $\mathcal{S}$, we construct the geometrical object known as the half-plane ($H_{l,A}$). This transition allows us to treat regions of the plane as sets that can be intersected to form more complex figures, such as angles and triangles.

Definition 10 (Half-Plane)**.** *Let $l \in \mathbf{L}$ be a line and let $A \in \mathbf{P}$ be a reference point such that $A \notin l$. The half-plane detirmined by l and A, denoted $H_{l,A}$, is the subset of the point-set consistening of all points on the same side of l as A. That is,*

$$H_{l,A} \coloneqq \{X \in \mathbf{P} \setminus l \mid \mathcal{S}(A, X, l)\}.$$

randomideas:

Chapter 4

Angles

As opposed to being a measure, we define an angle as angle as a geometric figure, ie: a set of points. Specifically, an angle is the union of two rays that share a common vertex.

Definition 11 (Angle)**.** *An **angle** is the union of two rays $\overrightarrow{AO}$ and $\overrightarrow{OB}$ that share a common vertex O, denoted as follows*

$$\angle AOB \coloneqq \overrightarrow{AO} \cup \overrightarrow{OB}.$$

Using the half-planes defined previously, we have the tools neccessary to describe the interior and exterior of an angle.

Definition 12 (Interior of an Angle)**.** *For any three non-collinear points $A, B, C \in \mathbf{P}$, the **interior** of $\angle ABC$ is the set:*

$$Int(\angle ABC) \coloneqq \{P \in \mathbf{P} \mid \mathcal{S}(P, A, \overleftrightarrow{BC}) \wedge \mathcal{S}(P, C, \overleftrightarrow{BA})\}.$$

Equivalently, this is the intersection of two half-planes:

$$Int(\angle ABC) \coloneqq H_{\overleftrightarrow{BC},A} \cap H_{\overleftrightarrow{BA},C}.$$

Definition 13 (Interior of an Angle)**.** *For any three non-collinear points $A, B, C \in \mathbf{P}$, the **interior** of $\angle ABC$ is the set:*

$$Int(\angle ABC) \coloneqq \{P \in \mathbf{P} \mid \mathcal{S}(P, A, \overleftrightarrow{BC}) \wedge \mathcal{S}(P, C, \overleftrightarrow{BA})\}.$$

Equivalently, this is the intersection of two half-planes:

$$Int(\angle ABC) \coloneqq H_{\overleftrightarrow{BC},A} \cap H_{\overleftrightarrow{BA},C}.$$

Chapter 5

Triangles

5.1 Defintion

define as the union of three line segments. the three segments must be unique.

define inside and outside of the triangle based on if a point x is on the same side of each comprising segment or not

Axiom 16 (SSS Congruency)**.** *Given two triangles $\triangle ABC$ and $\triangle XYZ$, if all sides of one are congruent to a respective side of the other triangle, the triangles themselves are considered congruent:*

$$\triangle ABC \cong \triangle XYZ.$$

For congruent triangles, corresponding parts are congruent.

Chapter 6

Circles

Chapter 7

Quadrilaterals

motivation

7.1 Definition

A **quadrilateral is defined as the union of four unique noncollinear segments as follows:**

$$\square ABCD := AB \cup BC \cup CD \cup DA.$$

Each segment is a side of the quadrilateral, and each point A, B, C, D is a vertex. Sides that share a vertex, such as AB and BC are adjacent, otherwise they are called opposite, such as AB and CD.

Theorem 31 (Property of Quadrilateral Sides)**.** *Each side of $\square ABCD$ contains its two endpoints and infinitely many interior points (by density).*

Proof. By the infinite interior points theorem, any given segment has infinitely many points strictly between its endpoints. Because the sides of a quadrilateral are defined as the segments comprising it, the above result applies to sides as well. Addionally, a given segment is defined to contain its endpoints. Similarily, this applies to the sides of a quadrilateral. Thus, this proves the result. □

Theorem 32 (Adjacent sides intersect only at the common vertex)**.** $AB \cap BC = \{B\}$*, and similarly for the other three adjacent pairs.*

Proof. Suppose a point $X \neq B$ belonged to both AB and BC. By the definitions of segment and betweenness, this forces A, B, C to be collinear, contradicting the non-collinearity condition. □

7.2 Diagonals

Definition 14. *Given a quadrilateral $\square ABCD$, the segments AC and BD are the* ***diagonals*** *of $\square ABCD$.*

7.3 Trapezoid/Trapezium

We define a trapezoid as a quadrilateral with at least one pair of parallel sides. (In some traditions, a trapezoid has exactly one pair; we choose the inclusive definition, which places parallelograms as a special case of trapezoids.)

Definition 15. *A convex quadrilateral $\square ABCD$ is a trapezoid if and only if*

$$\overleftrightarrow{AB} \parallel \overleftrightarrow{CD} \text{ or } \overleftrightarrow{BC} \parallel \overleftrightarrow{DA}.$$

7.3.1 Diagonals

7.3.2 Isosceles Trapezoid

A special case of the trapezoid occurs when the non-parallel sides (legs) are congruent. If this is the case, the quadrilateral is an isosceles trapezoid.

Definition 16 (Isosceles Trapezoid). *A trapezoid* $\square ABCD$ *with* $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$ *a isosceles trapezoid if and only if*

$$AD \cong BC.$$

Chapter 8

Quadrilaterals

motivation

8.1 Definition

8.2 Diagonals

Definition 17. *Given a quadrilateral* $\square ABCD$*, the segments* AC *and* BD *are the* ***diagonals*** *of* $\square ABCD$*.*

8.3 Trapezoid/Trapezium

We define a trapezoid as a quadrilateral with at least one pair of parallel sides. (In some traditions, a trapezoid has exactly one pair; we choose the inclusive definition, which places parallelograms as a special case of trapezoids.)

Definition 18. *A convex quadrilateral* $\square ABCD$ *is a trapezoid if and only if*

$$\overleftrightarrow{AB} \parallel \overleftrightarrow{CD} \text{ or } \overleftrightarrow{BC} \parallel \overleftrightarrow{DA}.$$

8.3.1 Diagonal

8.3.2 Isosceles Trapezoid

A special case of the trapezoid occurs when the non-parallel sides (legs) are congruent. If this is the case, the quadrilateral is an isosceles trapezoid.

Definition 19 (Isosceles Trapezoid)**.** *A trapezoid* $\square ABCD$ *with* $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$ *a isosceles trapezoid if and only if*

$$AD \cong BC.$$

8.3.3 Right Trapezoid

Definition 20 (Right trapezoid)**.** *A quadrilateral* $ABCD$ *is a* right trapezoid *if and only if it is a trapezoid and at least one of its interior angles is a right angle (if and only if at least one of its legs is perpendicular to the bases.)*

8.4 parallelograms

Theorem 33. *The diagonals of an isosceles trapezoid are congruent.*

Theorem 34 (Diagonals of isosceles trapezoid are congruent). *Given and isosceles trapezoid $\square ABCD$ with $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$, the diagonals are congruent:*

$$AC \cong BD.$$

8.5 Special Parallelograms

8.5.1 Rectangles

8.5.2 Rhombus

Definition 21 (Rhombus is a parallelogram whose sides are all the same length). *A rhombus is a parallelogram whose four sides are congruent to one another.*

Definition 22 (Rhombus is Equilateral but not Equiangular). *me e9tuer Rhombus is Equilateral but not Equiangular*

8.5.3 Square

8.6 Parallelograms

Parallelogram A quadrilateral is a parallelogram if both pairs of opposite sides are parallel.

Definition 23 (Parallelogram). *Let $A, B, C, D \in \mathbf{P}$ be four points satisfying the quadrilateral conditions. Then the quadrilateral $\square ABCD$ is a **parallelogram** if and only if*

$$\overleftrightarrow{AB} \parallel \overleftrightarrow{CD} \text{ and } \overleftrightarrow{BC} \parallel \overleftrightarrow{DA}.$$

i hate this project lmao ts straight retarded. idk what to do with it ever.

8.7 Special Parallelograms

8.7.1 Rectangles

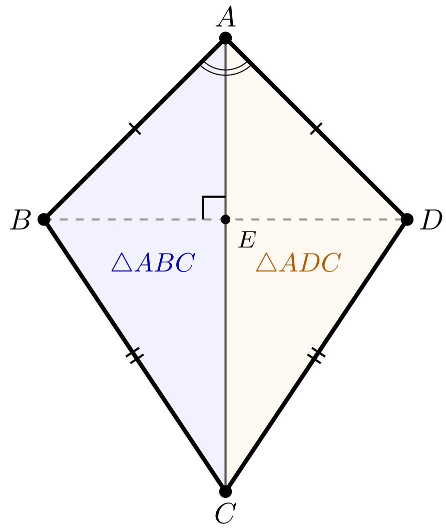
8.7.2 Rhombus

8.7.3 Square

8.8 Kites

Definition 24 (Kite). *A convex quadrilateral $\square ABCD$ is a **kite** if*

$$AB \cong AD \wedge CB \cong CD.$$



Theorem 35. *In a kite $\square ABCD$,*

$$\angle BAC \cong \angle DAC \text{ and } \angle BCA \cong \angle DCA.$$

Proof. Consider the triangles $\triangle ABC$ and $\triangle ADC$.

- $AB \cong AD$ by definition
- $CB \cong CD$ by definition
- $AC \cong AC$ by reflexivity of congruence

By the SSS triangle congruency criteria, $\triangle ABC \cong \triangle ADC$. Consequently, the corresponding angles are congruent:

$$\angle BAC \cong \angle DAC \text{ and } \angle BCA \cong \angle DCA.$$

□

Theorem 36 (The axis diagonal is the perpendicular bisector of the cross diagonal)**.** *In a kite $\square ABCD$, let $E = AC \cup BD$. Then,*

$$AC \perp BD \text{ and } BE \cong ED.$$

Proof. wasd □

Theorem 37 (Midpoint quadrilateral of a kite)**.** *The quadrilateral formed by joining the midpoints of the sides of a kite is a rectangle.*

Proof. □

8.8.1 Right Kite

8.9 Midpoint Theorems

8.10 Cyclic Quadrilaterals

8.11 Tangential Quadrilaterals

8.12 Bicentric Quadrilaterals

8.13 Other

11.15 Other Niche Quadrilaterals (Optional / Encyclopaedic) 11.15.1 Equidiagonal quadrilaterals (diagonals equal in length).

11.15.2 Orthodiagonal quadrilaterals (diagonals perpendicular).

11.15.3 Harmonic quadrilaterals (vertices form a harmonic division on the circumcircle – requires cross-ratio, so likely beyond your current primitives, but can be mentioned).

11.15.4 Quadrilaterals with special area properties (e.g., Brahmagupta's formula for cyclic quadrilaterals).

11.15.5 Complete quadrilaterals (the figure formed by four lines, six intersection points – a projective concept; may be out of scope).

11.15.6 Miquel point of a complete quadrilateral (four circles concurrent).

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